

Fig. 8: Multiplexing with GPIOs using 8x8 dot-matrix display

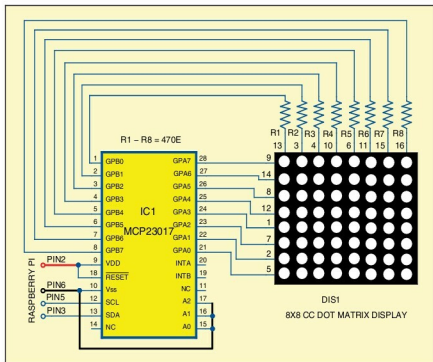


Fig. 9: Circuit of 8x8 dot-matrix display

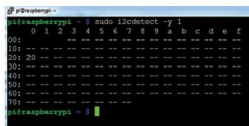


Fig. 10: Testing I2C connection

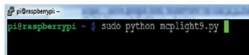


Fig. 11: Running the Python code

of figures on the 8x8 dot-matrix (using '1's) and the other to control the GPIOs for drawing figures on the LED matrix. So whatever letter or character you want to display, make

the changes in letter.py code and the same will be drawn on the LED matrix.

Make the connections as per the circuit shown in Fig. 9. MCP23017 (28-pin DIP) is used here as it has 16 GPIOs—8 for rows and 8 for columns of 8x8 LED matrix. Remember to use current-limiting resistors, whose values can be between 470 ohms and 1-kilo-ohm, in series with the LEDs.

Four wires are connected to the Raspberry Pi—two for I2C and two for 5V supply and ground connections.

Before testing, make sure that 5V supply is going to the correct pin on Raspberry Pi board, else it will heat

up the microprocessor and damage it. Also ensure the reset pin 18 of MCP23017 is connected with +5V rail.

Circuit and working

To test the I2C connection on Raspberry Pi, open the terminal and issue the following command:

```
$ sudo i2cdetect -y 0
```

or

```
$ sudo i2cdetect -y 1
```

If your Raspberry Pi is version 1 then use 0, else 1, and watch the screen shown in Fig. 10 appear at the terminal prompt.

Congratulations! Your Raspberry Pi has successfully detected the hardware at 0x20 address. The address is 0x20 because all the three address pins (A0, A1, A2) of MCP23017 are set low. If you had set only A0 high (connected to +5V) the address would have become 0x21.

This signifies one fundamental issue with I2C communications: You can connect any number of devices on the I2C bus (pin 3 and pin 5) but all devices should have unique addresses. So, setting A0, A1 and A2 pins will define different addresses for different devices. You can also control the device by issuing appropriate command. You may try this as explained below to gain more confidence.

There is another way of testing the I2C connection on Raspberry Pi. Connect an LED on A0 pin and issue the following commands:

```
$ sudo i2cset 0x20 0x00 0x00 # this
```

will switch on the LED on A0

```
$ sudo i2cset 0x20 0x00 0x01 # this
```

will switch off the LED on A0

The command line program is easy, provided you understand the binary and hex conversion. Adafruit.com has an excellent library using which we can manipulate these GPIOs in a far more lucid way. The files include Adafruit_I2C.py, Adafruit_MCP230xx.py, letter.py and mcpilight9.py

Now open the terminal and write the command given below (also refer Fig. 11):